

D781

Software Quality Assurance and Deployment

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A team has 900 end-to-end UI tests and 40 unit tests. The suite takes three hours and fails at random. The team uses a popular cloud CI service. What is the best way to rebalance?

- A. Move most checks into fast unit tests, keep some integration tests, and trim UI tests to key user journeys.
- B. Switch CI vendors.
- C. Delete all UI tests.
- D. Add more UI tests for coverage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

After a six-line fix to the tax calculation, a team ships without rerunning its older tests, and the shipping-cost feature breaks in production. Which practice was missing?

- A. Customer acceptance testing.
- B. None; a six-line change is too small to break anything.
- C. Automated regression testing on every change.
- D. Unit-testing only the six changed lines.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An age field accepts values from 18 to 65 inclusive. The form was designed by an outside agency and is offered in three languages. Using two-value boundary value analysis, which test values should be used?

- A. 0 and 100
- B. 18 and 65 only
- C. 30, 40, 50
- D. 17, 18, 65, 66

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A tester's boundary values are 0, 1, 250, and 251. What valid range is being tested?

- A.** 1 to 251
- B.** 1 to 250
- C.** 0 to 251
- D.** 0 to 250

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A store's discount rule: orders under \$50 get no discount; \$50 to \$199.99 get 10%; \$200 and up get 20%. About 40% of customers are loyalty members. Under equivalence partitioning, what is the minimum number of test values to cover each valid partition once?

- A.** Every possible dollar amount
- B.** 1
- C.** 3
- D.** 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Developers on a 15-person team merge to the main branch about once a month, and each merge takes days to untangle. The team uses Git. What does continuous integration prescribe?

- A.** Merge small changes at least daily, with an automated build and tests on every merge.
- B.** Automate deployment to production.
- C.** Switch away from Git.
- D.** Merge less often to reduce disruptions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Every change that passes the pipeline goes to production automatically, with no manual approval. What is this called?

- A.** Continuous integration.
- B.** Continuous deployment.
- C.** A scheduled quarterly release.
- D.** Continuous delivery.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bank with 2 million customers wants to release a new mobile API to 5% of users first, watch error rates, and then widen the rollout. Which strategy is this?

- A. Rolling update across servers.
- B. Blue-green deployment.
- C. Big-bang release to all users.
- D. Canary release.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team keeps two identical production environments. It switches all traffic from the old one to the new one at once and keeps the old one idle for instant rollback. Which strategy is this?

- A. In-place deployment with no rollback copy.
- B. Blue-green deployment.
- C. Rolling update.
- D. Canary release.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A database is backed up nightly at 1 a.m. It fails at 4 p.m., and a restore takes 3 hours. The data center is rated Tier III. How much data could be lost, and does that meet a 4-hour RPO?

- A. None; a Tier III data center prevents data loss.
- B. 3 hours; it meets the RPO.
- C. Up to 15 hours of data; it does not meet a 4-hour RPO.
- D. 15 hours, but it meets the RPO because the restore is under 4 hours.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A system's required RPO is 30 minutes. How often, at minimum, must backups or replication capture data?

- A. At least every 30 minutes.
- B. Every 60 minutes.
- C. It depends only on RTO.
- D. Nightly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Before a big ticket sale, a team runs the site at the expected peak of 20,000 users for an hour and checks response times. Next it wants to find the user count at which the site fails. The site uses a CDN. What is the second test called?

- A. CDN cache testing.
- B. Load testing.
- C. Soak testing.
- D. Stress testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A team has 900 end-to-end UI tests and 40 unit tests. The suite takes three hours and fails at random. ~~The team uses a popular cloud CI service.~~ What is the best way to rebalance?

Target word: rebalance the suite **Decides:** The pyramid is inverted.

A	Correct	Restores the pyramid: fast, stable feedback.
B	Camouflage	The vendor does not decide the test mix.
C	Absolute language	A few end-to-end checks are still valuable.
D	Inverted operation	Deepens the slow, flaky top layer.

2. Answer C Concept 2

After a ~~six-line~~ fix to the tax calculation, a team ships without rerunning its older tests, and the shipping-cost feature breaks in production. Which practice was missing?

Target word: missing practice **Decides:** A change broke an unchanged feature.

A	Adjacent category	Checks fitness for use, not regressions in each build.
B	Camouflage	Change size does not limit impact.
C	Correct	Rerunning existing tests catches side effects.
D	Dropped element	Misses effects on other features.

3. Answer D Concept 3

An age field accepts values from 18 to 65 inclusive. ~~The form was designed by an outside agency and is offered in three languages.~~ Using two-value boundary value analysis, which test values should be used?

Target word: boundary values **Decides:** Boundaries at 18 and 65.

A	Confusion set	Extreme values, not boundaries.
B	Dropped element	Drops the just-outside values.
C	Adjacent category	Mid-range values from one valid partition.
D	Correct	At and just outside each boundary.

4. Answer B Concept 3 · transfer

A tester's boundary values are 0, 1, 250, and 251. What valid range is being tested?

Target word: range — reversed from tests **Decides:** Valid edges are 1 and 250.

A	Wrong stem number	Takes 251 as valid.
B	Correct	0 and 251 are the just-outside values.
C	Inverted operation	Treats the invalid values as the edges.
D	Confusion set	Takes 0 as valid.

5. Answer C Concept 4

A store's discount rule: orders under \$50 get no discount; \$50 to \$199.99 get 10%; \$200 and up get 20%.

~~About 40% of customers are loyalty members.~~ Under equivalence partitioning, what is the minimum number of test values to cover each valid partition once?

Target word: count partitions **Decides:** Three valid discount classes.

A	Absolute language	Defeats the purpose of partitioning.
B	Dropped element	Covers one partition only.
C	Correct	One value per partition.
D	Adjacent category	Counts boundary pairs, a different technique.

6. Answer A Concept 5

Developers on a ~~15-person~~ team merge to the main branch about once a month, and each merge takes days to untangle. ~~The team uses Git.~~ What does continuous integration prescribe?

Target word: CI practice **Decides:** Infrequent, large merges.

A	Correct	Small, frequent merges keep conflicts small.
B	Adjacent category	That is CD; it does not fix merge pain.
C	Camouflage	The tool is not the cause.
D	Inverted operation	Larger merges cause bigger conflicts.

7. Answer B Concept 5 · transfer

Every change that passes the pipeline goes to production automatically, with no manual approval. What is this called?

Target word: name — reversed **Decides:** No human release step.

A	Adjacent category	Covers merging, building, and testing, not release.
B	Correct	Automatic release to production.
C	Inverted operation	The opposite of automatic release.
D	Confusion set	Keeps builds releasable but releases on a manual decision.

8. Answer D Concept 6

A bank ~~with 2 million customers~~ wants to release a new mobile API to 5% of users first, watch error rates, and then widen the rollout. Which strategy is this?

Target word: strategy name **Decides:** A small share of traffic first.

A	Adjacent category	Replaces servers in batches but is not a targeted user slice with a watch step.
B	Confusion set	Switches all traffic at once.
C	Inverted operation	Everyone at once.
D	Correct	A small slice of traffic tests the new version.

9. Answer B Concept 6 · transfer

A team keeps two identical production environments. It switches all traffic from the old one to the new one at once and keeps the old one idle for instant rollback. Which strategy is this?

Target word: strategy — reversed **Decides:** Two environments, full switch.

A	Dropped element	Drops the idle environment.
B	Correct	Two environments with a full traffic switch.
C	Adjacent category	Replaces servers gradually in one environment.
D	Confusion set	Routes only a small share first.

10. Answer C Concept 7

A database is backed up nightly at 1 a.m. It fails at 4 p.m., and a restore takes 3 hours. ~~The data center is rated Tier III.~~ How much data could be lost, and does that meet a 4-hour RPO?

Target word: data loss vs. RPO **Decides:** Last good copy is 15 hours old.

A	Camouflage	Facility tier does not replace backups.
B	Confusion set	3 hours is restore time (RTO).
C	Correct	1 a.m. to 4 p.m. is 15 hours of changes.
D	Adjacent category	Restore speed is RTO, not RPO.

11. Answer A Concept 7 · transfer

A system's required RPO is 30 minutes. How often, at minimum, must backups or replication capture data?

Target word: solve for backup frequency **Decides:** The gap between copies cannot exceed the RPO.

A	Correct	Any longer gap could lose more than 30 minutes.
B	Inverted operation	Doubles the allowed gap.
C	Confusion set	RTO governs restore time, not backup frequency.
D	Dropped element	Ignores the RPO.

12. Answer D Concept 8

Before a big ticket sale, a team runs the site at the expected peak of 20,000 users for an hour and checks response times. Next it wants to find the user count at which the site fails. ~~The site uses a CDN.~~ What is the second test called?

Target word: test type **Decides:** Pushing past peak to find the breaking point.

A	Camouflage	The CDN does not define the test type.
B	Confusion set	That was the first test, at expected peak.
C	Adjacent category	Runs a long time to find leaks.
D	Correct	Finds limits beyond expected load.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The test pyramid	Keep many fast unit tests, fewer integration tests, and few slow end-to-end UI tests.	An inverted pyramid dominated by slow, flaky UI tests.	A long, flaky suite made mostly of UI tests.
2	Regression testing	After any change, rerun existing tests (ideally automated in CI) to catch breaks in unchanged features.	Testing only the changed lines.	A small fix that broke a different feature.
3	Boundary value analysis	Test just below, at, and just above each boundary (two-value: at and just outside).	Testing only mid-range or only the boundaries themselves.	An inclusive numeric range.
4	Equivalence partitioning	Divide inputs into classes that behave the same and test one value from each class.	Testing every value, or one value total.	Tiers or bands of input with different rules.
5	CI vs. continuous delivery vs. continuous deployment	CI merges small changes often with automated build and test; continuous delivery keeps every build releasable with a manual release step; continuous deployment releases automatically.	Mixing up delivery and deployment.	Whether a human approves the production release.
6	Deployment strategies	Canary sends a small share of traffic to the new version first; blue-green switches all traffic between two identical environments; rolling replaces servers in batches.	Confusing canary with blue-green.	"5% of users first" vs. "switch all traffic, keep old idle."
7	RPO vs. RTO	RPO = maximum acceptable data loss, measured back to the last good copy; RTO = maximum acceptable downtime to restore.	Swapping the two.	Backup time, failure time, and restore time.
8	Performance test types	Load testing checks expected peak; stress testing pushes past it to find the breaking point; soak testing runs long to find leaks.	Calling a breaking-point test a load test.	"Find the number of users at which it fails."

Course scope (public description): This course equips students with the skills and knowledge necessary for ensuring high-quality software development and effective deployment practices. The curriculum covers quality models and metrics, QA methods, automated testing, and the design and implementation of test cases. This course introduces students to continuous integration/continuous deployment (CI/CD) pipelines and tools, performance testing, and the tools required for assessing software performance. Students will also learn about deployment strategies, rollback procedures, disaster recovery plans, and monitoring and logging practices, each crucial for the software's operational integrity in production environments.